

# Social Media Engagement Data Analysis

## 1. Project Overview

This project executes an end-to-end Data Analysis Pipeline on a 5,000-row Social Media Engagement dataset. The primary objective is to clean raw data, perform exploratory data analysis, engineer new engagement features, compute descriptive statistics, and extract actionable insights to optimize social media performance.

## 2. Key Execution Steps Across 6 Tasks

### Data Acquisition & Preprocessing

- Loaded the raw dataset from GitHub into Python using Pandas.
- Imputed missing values across critical numerical metrics (likes, comments, shares) using median and mode imputation.
- Corrected negative values and formatted temporal attributes (posted\_at) into standard datetime objects (dayfirst=True).

### Feature Engineering & Text Cleaning

- Extracted hashtag frequency into a dedicated feature column (hashtag\_count) using custom string manipulation functions.
- Standardized categorical text data (sentiment, post\_type) into clean lowercase and title-case formats.

### Data Exploration (EDA)

- Inspected structural metadata using head(), tail(), shape, and info().
- Evaluated categorical distributions via value\_counts() and identified linear feature dependencies using a Pearson correlation matrix.

### Data Wrangling

- Merged post categories using relational key mappings (pd.merge).
- Engineered a composite performance score:  
$$\text{engagement\_score} = \text{likes} + (\text{comments} \times 2) + (\text{shares} \times 3)$$
- Applied logarithmic transformations (log\_likes) to normalize skewed metric distributions and generated grouped aggregations by platform, country, and sentiment.

### Statistical Analysis

- Computed central tendency (Mean, Median, Mode) and dispersion metrics (Standard Deviation, Variance, Percentiles: 25%, 50%, 75%) across key numerical fields (likes, comments, shares, watch\_time, followers).

- Evaluated distributional asymmetry and tail behavior using Skewness and Kurtosis.

### **Data Visualization & Strategic Insights**

- Built interactive visualizations (Histograms, Bar Charts, Box Plots, and Correlation Heatmaps) using Seaborn and Matplotlib.
- Identified top-performing content formats, demographic engagement levels across countries, and sentiment impact on audience reach.

### **3. Final Conclusion**

The analysis demonstrates that composite engagement metrics and hashtag frequency significantly correlate with organic reach. These findings provide a data-driven framework for content creators and marketers to refine posting strategies, target high-performing regions, and maximize audience interaction.